

Saipranav Telidevarapalli

US CITIZEN

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Education

Georgia Institute of Technology

BS/MS in Aerospace Engineering

Atlanta, GA

BS Dec 2026

MS Dec 2027

- **Relevant Coursework:** Engineering Materials, Air/Space Vehicle Performance, Thermodynamics and Fluids, Aerothermodynamics, Jet & Rocket Prop., Dynamics, Aerodynamics, Ctrl Sys. Analysis, Sys. Dynamics & Vibrations, Def. Bodies

Experience

Relativity Space

Long Beach, California

Mechanical Engineer Intern

(40 hours/week) May 2026 - Aug 2026

- Evaluated and employed experimental design to spearhead development of cold spray additive manufacturing process
 1. Rapidly developed, tested, and iterated on mechanical components to improve the cold spray processes
 2. Wrote a heat transfer sim to model temperature loss during spray process
- Led the design and CFD analysis to develop a mechanical component that reduces oxidation contamination on cold sprayed materials - Reduced oxygen pickup by 70 percent
- Optimized the cold spray nozzle using Method of Characteristics and CFD to increase particle velocities and reduce any flow losses

Georgia Tech Supersonics Club

Atlanta, Georgia

GNC and Propulsion Engineer

(6 hours/week) Jan 2024 - Present

- Implementing Extended Kalman Filter(EKF) to fuse navigation based sensor data and FDM predicted states for optimal state estimation
- Developing a full 6DOF flight dynamics model in Simulink using CFD-derived aerodynamic data
- Lead the design (CAD) and additive manufacturing choices of supersonic jet nozzles through CFD analysis leveraging Ansys Fluent
 1. Optimized plug and nozzle shape designs by observing total pressure, static pressure, and exit velocity contours
- Implemented a parametric-cycle-analysis script in MATLAB/Python to simulate chemical eq. ratios and thermodynamic properties throughout all sections of engine assemblies for characterizing engine properties(exit velocity, thrust, etc.)
- **Skills Used:** Solidworks, Ansys, Simulink, Python, Microsoft 365

Aerospace Robotics Laboratory

Atlanta, Georgia

GNC Engineer

(4 hrs/week) Aug 2025 - Present

- Working on the state-estimation and control architecture for a lunar terrain rover
 1. Using ROS2 framework to build an efficient feedback controller for the rover to run in various autonomous modes
 2. Integrating neural-network based data into an EKF to improve state estimation error caused by slipping dynamics on unexpected terrain conditions
- **Skills Used:** Physics, Engineering Analysis, python, Control and Dynamics

Yellow Jacket Space Program

Atlanta, Georgia

Propulsion and Structures Engineer

(4 hours/week) Aug 2023 - Present

- Designing the Vehicle structures for Elytra (a student rocket intended to reach space) using static and dynamic loads analysis
- Worked deeply with mechanical design, assembly, and test of a self-actuated launch lug system
 1. Used physics principles to perform static analysis, design, and build of the system as well as testing rig
- Heavily involved in designing team's space-shot vehicle's fluids design using P&ID diagrams
 1. Worked on GSE pressurized fluids lines to fill vehicle's propellants
 2. Added vent valves, isolation valves, and check valves to critical areas for better safety of the ground fluids system
- **Skills Used:** Solidworks, MATLAB, Ansys, FEA

Airbus

Mobile, Alabama

Systems Engineering Intern

(40 hrs/week) May 2025 - August 2025

- Worked on system modeling to develop models for cabin systems
 1. Developed a model to find critical insufficient links between Airbus' certification documents and EASA's documents
 2. Built a seats model from scratch to outline its life cycle from design and certification to production and commercial use

- Researched under Asst. Prof. Nandeesh Hiremath to explore the applications of Machine Learning in Aerospace engineering.
- Developed Python-based neural network for atmospheric object classification in security and defense applications
- Utilized OpenCV and TensorFlow for rapid data analysis of various shapes based on contours
- **Skills Used:** Tensorflow, OpenCV, Python

Projects

Computational Combustion Lab - Python Exascale CFD Solver

Atlanta, Georgia

CFD Software Developer

(10 hours/week) August 2024 - June 2025

- Designed a CFD solver for optimal performance on exascale super computing architectures by leveraging parallelism and cutting-edge hardware accelerators
- Scaled parallelized explicit schema code to create a foundational solver for 1D, 2D, and 3D convection problems
- **Skills Used:** Python, cuPy, mpi4Py